
D3: Hyperspectral (HRS) data analysis

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CONTENTS

CONTENTS	2
TABLES	3
FIGURES	4
1. INTRODUCTION	5
1.1 AIMS AND OBJECTIVES	6
1.2 STUDY AREA	7
2. METHODS AND MATERIALS.....	8
2.1 DATA PREPARATION AND PREPROCESSING.....	9
2.2 FEATURE SELECTION AND TRANSFORMATION	9
2.3 MACHINE LEARNING ALGORITHMS.....	10
2.3.1 RANDOM FOREST	10
2.3.2 EXTREME GRADIENT BOOSTING(XGBoost).....	10
2.4 HYPERPARAMETER OPTIMISATION.....	10
2.5 TEMPORAL TRANSFERABILITY	11
3. RESULTS AND DISCUSSION.....	11
3.1 RANDOM FOREST AND XGBOOST BASELINE.....	11
3.2 FEATURE SELECTION	15
3.2.1 BORUTA FEATURE SELECTION	15
3.2.2 RECURSIVE FEATURE ELIMINATION (RFE)	18
3.3 HYPERPARAMETER OPTIMISATION AND FEATURE TRANSFORMATION.	19
3.4 TEMPORAL TRANSFERABILITY	21
4. CONCLUSION	23
REFERENCES	24

TABLES

Table 1: F1-Score(weighted) and Matthews Correlation Coefficient for the two baseline models.	11
Table 2: F1-Score(weighted) and Matthews Correlation Coefficient for the two models using Boruta Feature selection.	16
Table 3: F1-Score(weighted) and Matthews Correlation Coefficient for the two models using Boruta Feature selection.	18
Table 4: F1 and MCC scores for the final model.	20

FIGURES

Figure 1: Study Area in Welgevallen Experimental Farm, Stellenbosch	7
Figure 2: Hyperspectral machine learning workflow for the discrimination of grapevine cultivar.	8
Figure 3: Full spectral dataset with grey zones highlighting areas of atmospheric water absorption.	9
Figure 4: ROC curves and AUC score per grapevine class using Random Forest.	12
Figure 5: ROC curves and AUC score per grapevine class using XGBoost.....	12
Figure 6: Precision, Recall, and F1-score per cultivar using the Random Forrest baseline.....	13
Figure 7: Precision, Recall, and F1-score per cultivar using the XGBoost baseline.	14
Figure 8: Selected Features using Boruta Wrapper feature selection method.	15
Figure 9: ROC curves and AUC scores of RF Boruta Feature selection.	17
Figure 10: ROC curves and AUC scores of XGB Boruta Feature selection.....	17
Figure 11: Selected Features using RFE Wrapper feature selection method.	18
Figure 12: ROC curves and AUC scores of XGBoost final model.....	20
Figure 13: ROC curve and AUC scores of the Temporal transferability test on January data.	22

1. INTRODUCTION

Viticulture is the study and practice of cultivating grapevines, this practice requires understanding of the plant phenotypes and their characteristics to allow for grapevine cultivar discrimination. Grapevine cultivar discrimination is pivotal for vineyard management to assess vineyard yield and plant health which equips farmers with the knowledge of what nutrients and water is needed. However, distinguishing between the different grape phenotypes is challenging for the grape growing and wine industry (Gutiérrez et al., 2018).

Distinguishing between the different phenotypes is historically done using destructive techniques like DNA analysis (Gutiérrez et al., 2018). Not only are these methods time intensive and costly they can only process data on a small level, this creates an urgent need for a fast, inexpensive, and scalable approach to grapevine cultivar discrimination. Remote sensing offers a non-destructive method for acquiring data needed to differentiate between different cultivars through the use of a combination hyperspectral data and machine learning ensembles.

The use of hyperspectral imagery and machine learning has been used to discriminate different common grapevine varieties in previous studies. Specifically, (Gutiérrez et al., 2018) aimed to classify 30 different grapevine varieties under field conditions using an on-the-go hyperspectral imagery system. The use of Support Vector Machines (SVM) and Multilayer perceptrons (MLP), the researchers achieved F1-scores up to 0.99. This demonstrates that these methods were effective in the classification of different grapevines and that machine learning methods can be applied to hyperspectral imagery.

High accuracies have been achieved using different ML algorithms, for example (Loggenberg et al., 2018), used two different ensemble learning algorithms Random Forest and XGBoost on hyperspectral data. The results gathered by this study concluded that for the modelling of water stress in a Shiraz vineyard that Random Forest outperformed XGBoost. While the analysis was applied to one cultivar of vineyard, this demonstrates the ability of Machine learning algorithms to be used accurately in various different applications.

Hyperspectral imagery provides a vast amount of spectral information as it contains reflectance from the visible, near infrared, and shortwave infrared regions of the electromagnetic spectrum (Loggenberg & Poona, 2022). These wavelengths are stored in narrow bands which allows for a continuous spectral signature, this contrasts the discrete nature of multispectral imagery. Hyperspectral imagery contains high dimensional data, feature selection techniques are used to reduce the data dimensionality and select subsets of features that are significant to create a model on (Loggenberg & Poona, 2022). Various methods can be used to identify important bands, these include filter wrapper and embedded methods (Chandrashekar & Sahin, 2014).

1.1 AIMS AND OBJECTIVES

This study aims to develop and assess a machine learning and remote sensing framework used for grapevine cultivar discrimination and the temporal transferability. Where different ML classifiers such as Random Forests (RF) and XGBoost will be used on hyperspectral data with different feature selection and feature transformation techniques applied to it. Where similar studies, for example (Gutiérrez et al., 2018), aimed to classify various different cultivar of grapevines. This study will focus on the following raisin grapevines; Currants, Merbein Seedless, Diamond Muscat, Selma Pete, Sugra-39, and Sultana.

To achieve this aim, the following objectives will be addressed:

- Preprocessing of vineyard leaf hyperspectral data to remove duplicate samples and remove atmospheric absorption bands.
- Implement and assess Random Forest and XGBoost classifiers on the pre-processed hyperspectral data.
- Implement Boruta and Recursive Feature Elimination and assess how the selected features impacted the machine learning classifier's results.
- Perform feature transformations and hyper parameter optimisation to assess the accuracy and performance of the model using those features.
- Evaluate and assess the temporal transferability using data that was acquired one month after the training data.

1.2 STUDY AREA

Study Area

Welgevallen Experimental Farm, Stellenbosch.



Figure 1: Study Area in Welgevallen Experimental Farm, Stellenbosch

2. METHODS AND MATERIALS

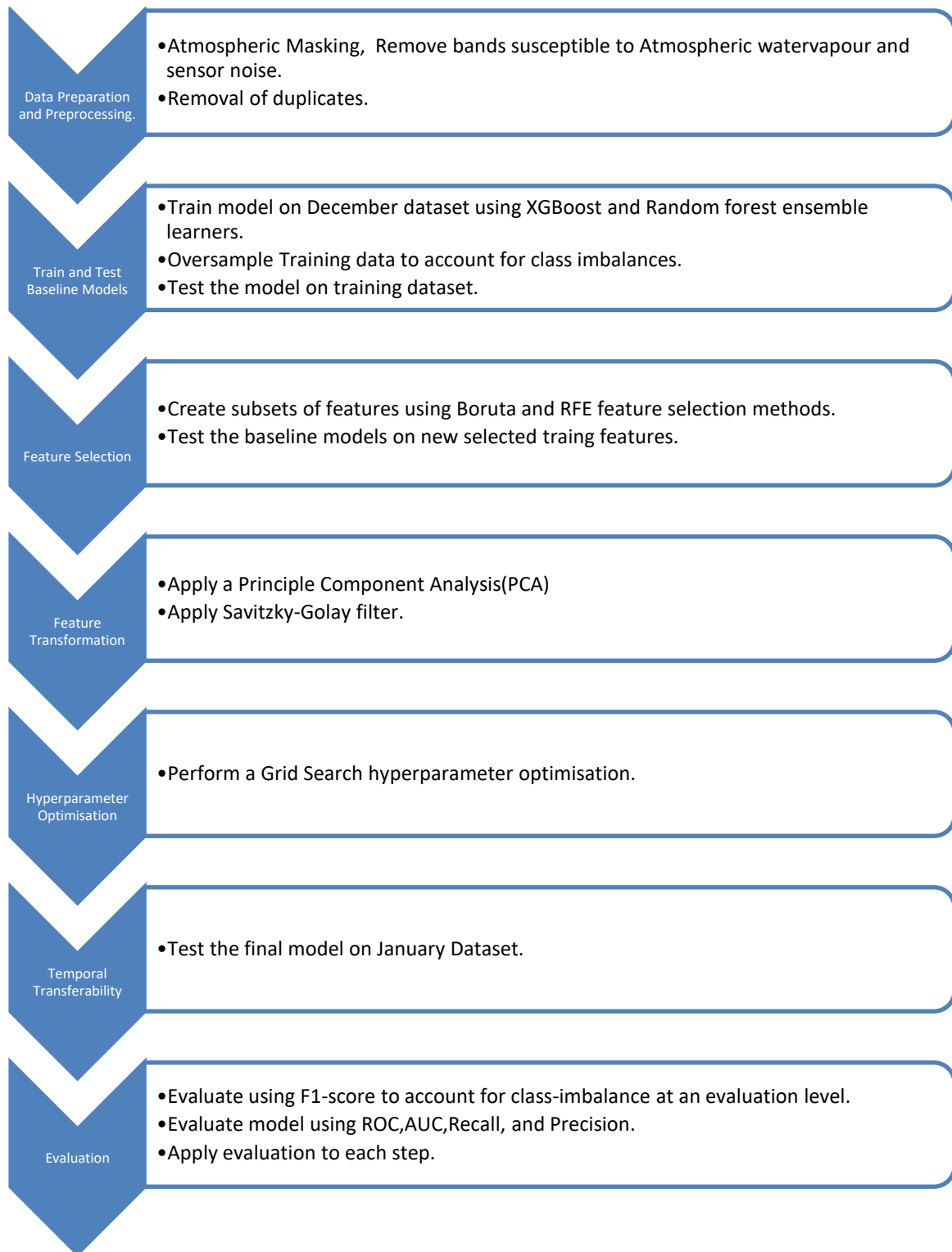


Figure 2: Hyperspectral machine learning workflow for the discrimination of grapevine cultivar.

2.1 DATA PREPARATION AND PREPROCESSING

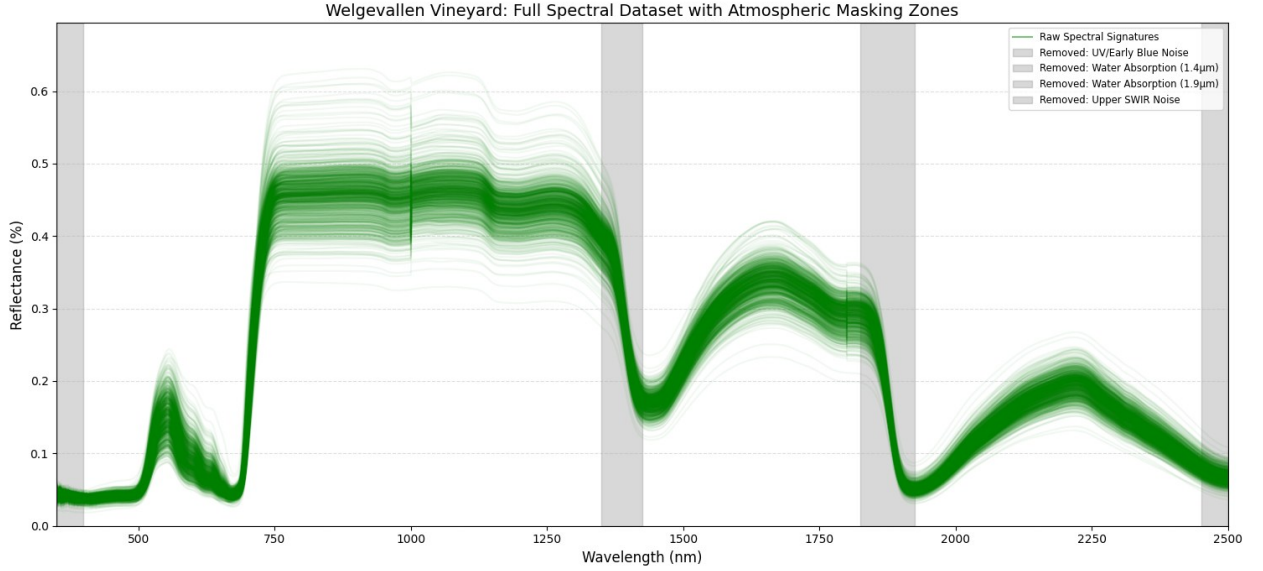


Figure 3: Full spectral dataset with grey zones highlighting areas of atmospheric water absorption.

The main objective when preprocessing the hyperspectral data is to remove the collection of wavelengths in known atmospheric windows. Previous literature indicates that these bands of atmospheric water absorption occurred roughly between $1.05\mu\text{m}$ to $1.1\mu\text{m}$, $1.3\mu\text{m}$ to $1.5\mu\text{m}$, $1.7\mu\text{m}$ to $2.1\mu\text{m}$ (Gao et al., 1993). This aligns with the grey bands illustrated in Figure 3, where wavelengths 350 – 399 nm, 1350 – 1425 nm, 1825 – 1925 nm, and 2451 – 2500 nm were omitted, this left 1874 bands for future feature selection.

2.2 FEATURE SELECTION AND TRANSFORMATION

To overcome the “curse of dimensionality”(Hughes Phenomenon), where a large set of spectral bands lead to model overfitting, feature selection and feature transformation is required.

Feature selection is the isolation of significant features for a model to be trained and tested on. This study uses a Boruta selection and a Recursive Feature Elimination (RFE). Boruta selection is a wrapper method that uses real bands to shadow over features, bands are selected depending on significance to the model. Boruta also includes bands that are highly correlated. Recursive feature elimination uses a minimal-optimal approach that removes unimportant bands iteratively. This method used a subset of 115 features, this was chosen to reduce redundancy. These feature selection methods were first applied to the Random Forest model, as this reduced computational time and allowed both ensemble learners to be run on the same band subsets which creates an a

fair comparison. This approach was used in previous studies (Löw et al., 2013), where Random Forest was used for feature selection due to its robustness against overfitting and the computation of feature importance, this feature selection was then used on a Support Vector Machines model.

Feature transformation is the process of using mathematical formulas to turn raw data into more useful variables. This can reduce dimensionality and remove bands that have high collinearity. Principle Component Analysis (PCA) is used to transform 1874 highly correlated bands into a subset of 51 uncorrelated bands. The subset of 51 features for the PCA was found to be effective on the performance of the model through iterative testing.

2.3 MACHINE LEARNING ALGORITHMS

2.3.1 RANDOM FOREST

Random Forest is an ensemble classifier that combines a number of decision trees to produce a single prediction, each tree is grown using a bootstrap sample which makes each tree unique (Breiman, 2001). This method means that not every wavelength is used to train the model, rather a set of randomly sampled bands. Random Forest is robust to noise makes it potentially ideal for hyperspectral datasets which are prone to noise. To create a baseline Random Forest classifier the Scikit-learn RandomForestClassifier was used in python with default hyperparameters. The Random Forest model was trained on 0.66 of the data where the remaining 0.33 was used to test the model. After the model is trained a method of upsampling is used to account for the class imbalance at a data-level.

2.3.2 EXTREME GRADIENT BOOSTING(XGBoost)

XGBoost is an ensemble classifier that uses an advanced boosting method that simultaneously examines multiple variables to create predictions (Chen & Guestrin, 2016). XGBoost uses an iterative approach, where decision trees are grown on a boosting sample. The XGBoost classifier was trained on 0.66 of the data and the remaining 0.33 was used to test the model. To create a baseline, the XGBoost used default hyperparameters.

2.4 HYPERPARAMETER OPTIMISATION

Hyperparameters are the ruleset of the classifier, these rulesets include maximum depth of a tree, learning rate, and number of estimators. In this study a Gridsearch method will be used to test combinations of number of estimators and the depth of a tree.

2.5 TEMPORAL TRANSFERABILITY

To assess the temporal transferability the best performing model containing different feature selection and feature transformations will be tested on the January data. This temporal transferability will be an indicator of overfitting within the model.

3. RESULTS AND DISCUSSION

3.1 RANDOM FOREST AND XGBOOST BASELINE

The Random Forest and XGBoost baseline models were compared against each other using different statistical metrics to determine which classifier performed the best at discriminating the different grapevine cultivar.

Table 1: F1-Score(weighted) and Matthews Correlation Coefficient for the two baseline models.

	RF Baseline	XGB Baseline
F1-Score (Weighted)	0.550972	0.610317
MCC	0.462441	0.539028

In Table 1, the XGBoost baseline achieved an F1-Score of 0.61 which is significantly larger than the F1-score of the Random Forest baseline. Both models exhibit low F1-scores, this is a result of the models having low precision and recall of the various grapevine classes. The XGBoost baseline achieved a MCC score of 0.54 while this value is low it is an improvement over the Random Forest Baseline score of 0.46. This difference in performance is likely attributed to the iterative and internal cross validation of the Xgboost which minimises residual rather than the parallel ensemble method of the Random Forest. Both models exhibited moderate positive MCC values, which indicates the classification of the cultivar types is not randomly assigned.

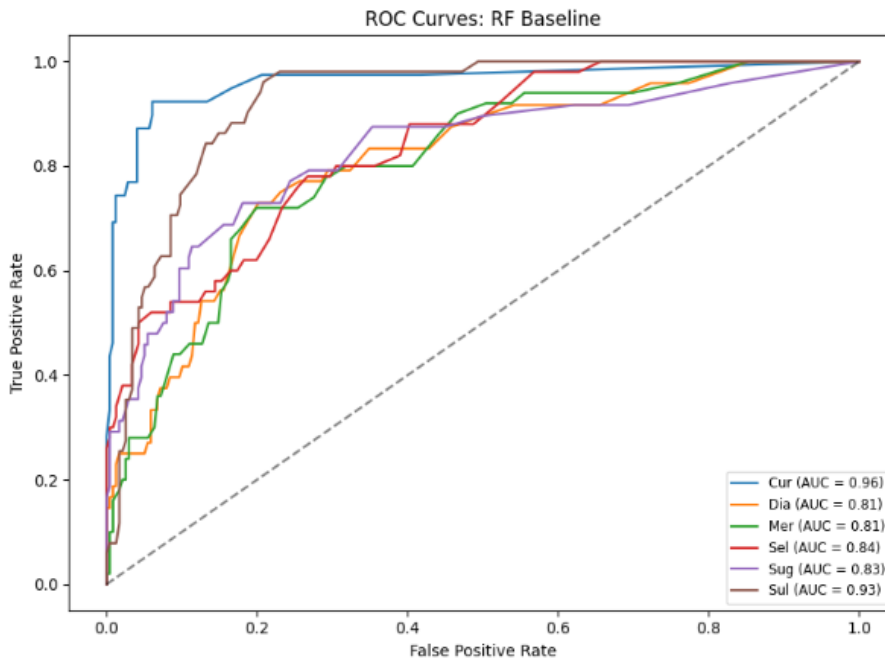


Figure 4: ROC curves and AUC score per grapevine class using Random Forest.

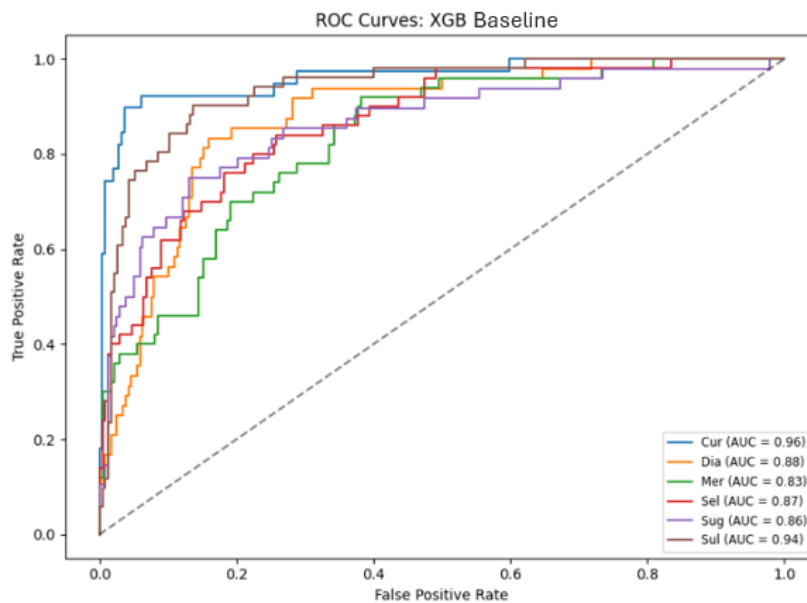


Figure 5: ROC curves and AUC score per grapevine class using XGBoost.

In Figure 4, the ROC curves of the Random Forest baseline bend to the top left and away from the AUC baseline score of 0.5. This is the optimal area for ROC curves to lie as it indicates a larger true positive rate with low false positives. The curves in Figure 4 indicate that while there is a discrimination between the classes multiple classes such as Merbein and Diamond Muscat lie closer to the baseline this indicates that there is a spectral similarity when discriminating those cultivar from others.

In Figure 5, the ROC curves of the XGBoost baseline are similar to that of the Random Forest as they bend to the top left but more of the cultivars that were discriminated by the XGBoost contain a higher true positive rate. The XGBoost baseline model exhibited high AUC in all cultivar classes which is a significant improvement over the Random Forest model, where classes such as Diamond Muscat contained more false positives in the RF model than the XGBoost model.

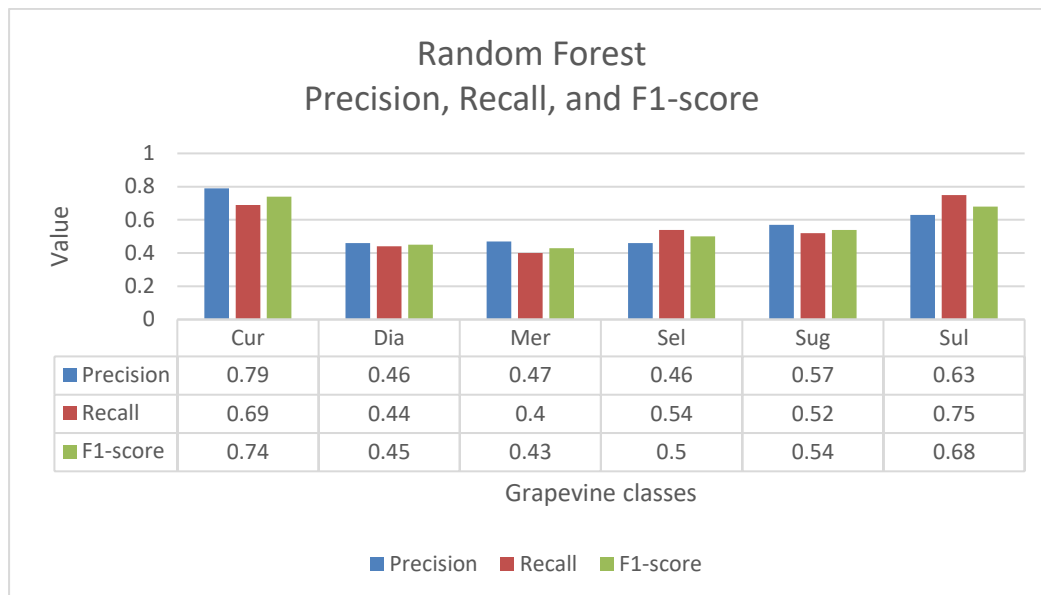


Figure 6: Precision, Recall, and F1-score per cultivar using the Random Forrest baseline.

Figure 6 indicates that Sultana and Current grapevine cultivar exhibited higher recall and precision than the other classes. The precision of the Current cultivar was substantially larger than the recall, this results in more instances of the model predicted as Current was actually classed as current. The other cultivar recorded recall and precision values in the range of 0.4 to 0.57, this indicates that these classes may have similar spectral qualities which at a baseline level proves ineffective at discriminating between different cultivar.

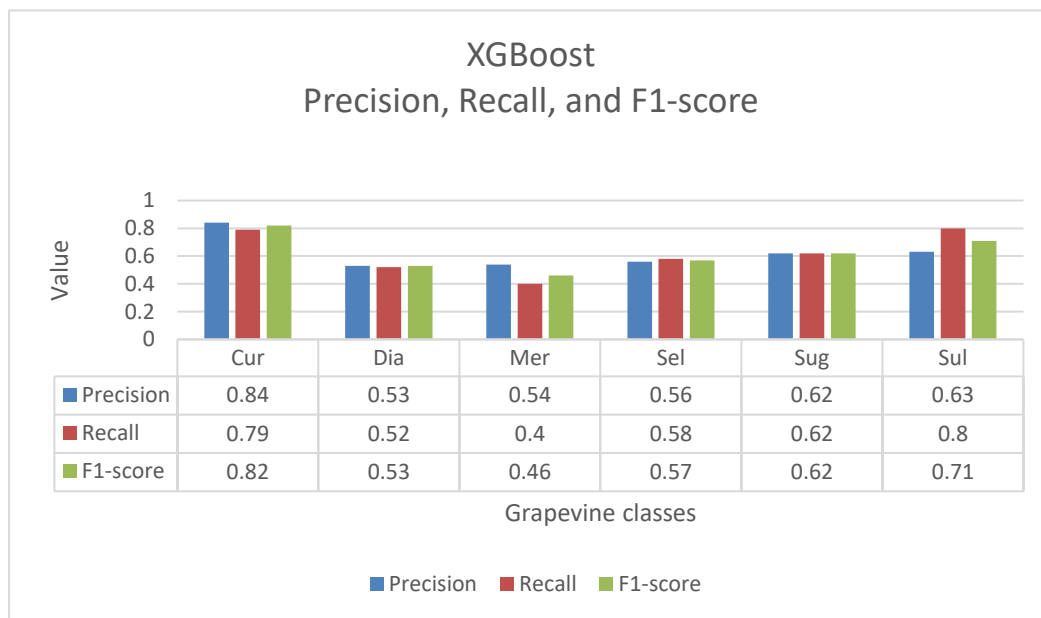


Figure 7: Precision, Recall, and F1-score per cultivar using the XGBoost baseline.

Figure 7 indicates that the Current and Sultana cultivar were able to be classified with more precision and recall than the other cultivar with a F1- score of 0.82 and 0.71 respectively. The F1-scores of the other cultivar range between 0.46 and 0.62 which is a significant increase from the RF baseline.

Based on the baseline results, it is evident that XGBoost outperformed the Random Forest model. This outcome was unexpected, as the Random Forest classifier is regarded as robust to noise which is a significant factor in this baseline dataset, which contained a large number of features with minimal feature selection. At the class level, the Currant and Sultana cultivars were the most successfully discriminated, likely due to these varieties exhibiting unique spectral characteristics. This high performance in the two leading classes inflated the weighted average F1-score seen in Table 1. The superior performance of XGBoost indicates that its boosting approach is more effective at internal validation and distinguishing between classes in this context, where the Random Forest approach proved less successful.

3.2 FEATURE SELECTION

3.2.1 BORUTA FEATURE SELECTION

The Boruta feature selection wrapper was implemented using a Random Forest classifier to identify a subset of significant spectral bands. This reduced feature set was used to evaluate both the Random Forest and XGBoost models, allowing for a comparison against the high dimensional baseline results.

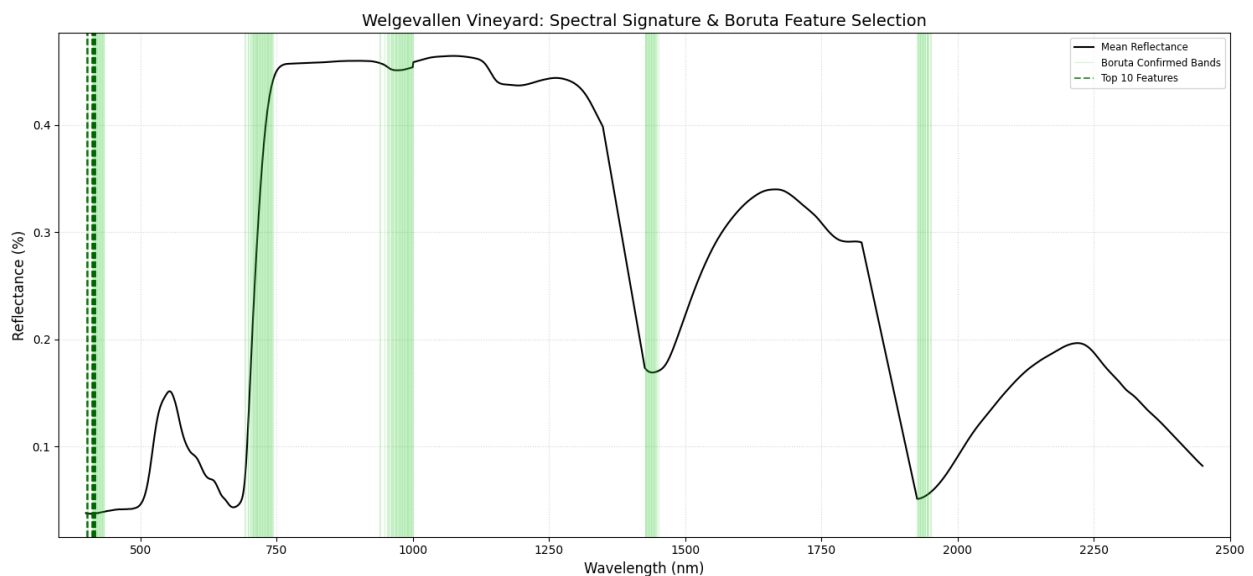


Figure 8: Selected Features using Boruta Wrapper feature selection method.

Figure 8 illustrates the features selected by the Boruta feature selection method alongside the top 10 most significant features within the subset. These selections align with absorption troughs in the mean spectral signature of grapevines experiences a decrease in percentage reflectance. The identified features span the Blue, Red, Red Edge, NIR, and SWIR regions. Wavelengths selected in the Blue and Red regions largely selected as these are areas where chlorophyll primarily absorbs light. Conversely, the green region was not selected as this is where chlorophyll reflects more light, which suggest that more information is contained in regions of absorption. Red-edge features were selected due to their sensitivity to chlorophyll concentration. The bands selected in the Near infrared region provide information of the cell structure of the leaves where shifts in this region will indicate the health of the leaf. Short wave Infrared was selected as the SWIR region indicates the water content of the grapevine where shifts or dips in this region indicate the change in water content of the leaves. Overall the Boruta method selected 163 features that were deemed significant.

Table 2: F1-Score(weighted) and Matthews Correlation Coefficient for the two models using Boruta Feature selection.

	RF Boruta	XGB Boruta
F1-Score (Weighted)	0,539279	0,592954
MCC	0,449733	0,517696

In Table 2, the XGBoost model achieved an F1-score of 0.59 which is larger than the RF model. Due to the decrease in dimensionality it is expected that the accuracies would slightly increase due to the reduction of noise, conversley, the XGBoost model tested on selected features from the boruta method decreased in overall performance when compared to the baseline assessment. Although the RF model exhibits a lower F1-score than the XGBoost, there is a slight increase in performance from the baseline. The MCC values of both classifiers have decreased from the baseline assessment, this results in a slight increase in the random assignments of classes to samples.

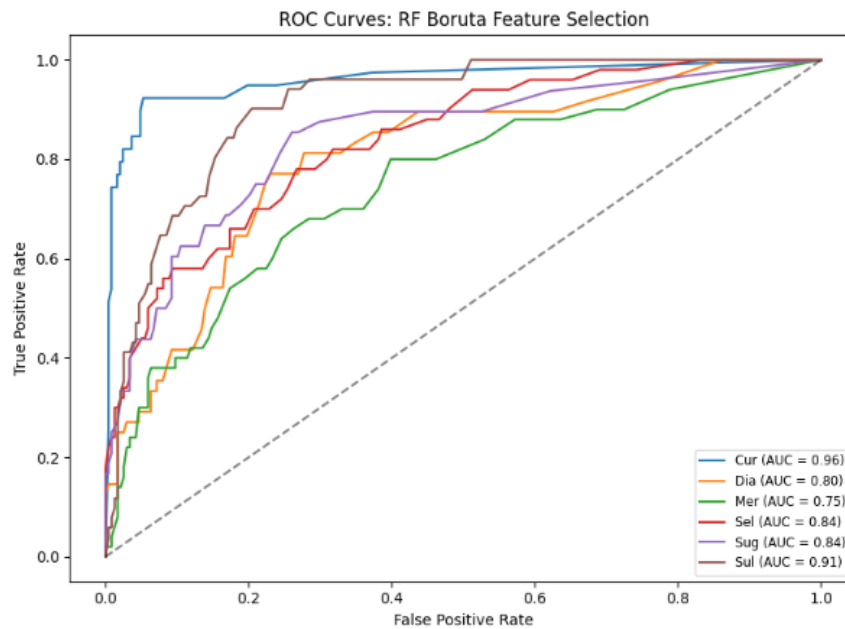


Figure 9: ROC curves and AUC scores of RF Boruta Feature selection.

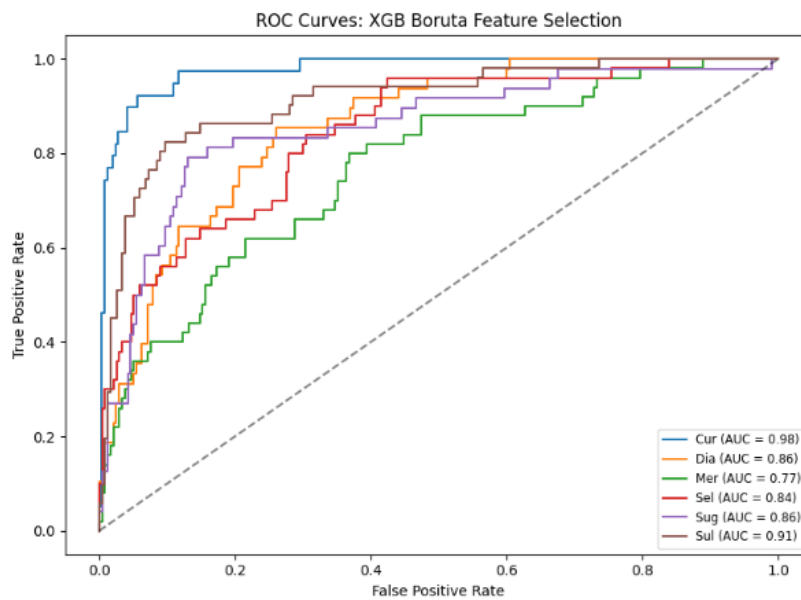


Figure 10: ROC curves and AUC scores of XGB Boruta Feature selection.

The addition of the Boruta feature selection increased certain class AUC scores as seen in the Currant cultivar increasing in the XGBoost after feature selection. The ROC curves roughly stayed similar to the base line assessment, seen in Figures 9 and 10. This variation in AUC scores between the baseline and feature selection models is negligible in the overall accuracy of the model.

In comparison to the baseline performance for both Random Forest and XGBoost, the models using the Boruta Feature selection exhibited a slight variation in the performance of the model. This highlights the effectiveness of the Boruta features selection to remove redundant bands without significantly impacting the performance of the model.

3.2.2 RECURSIVE FEATURE ELIMINATION (RFE)

RFE is a feature selection method that isolated 115 bands. The RFE was applied to the Random forest baseline model to identify the band selection to be applied in both XGBoost and Random Forest models.

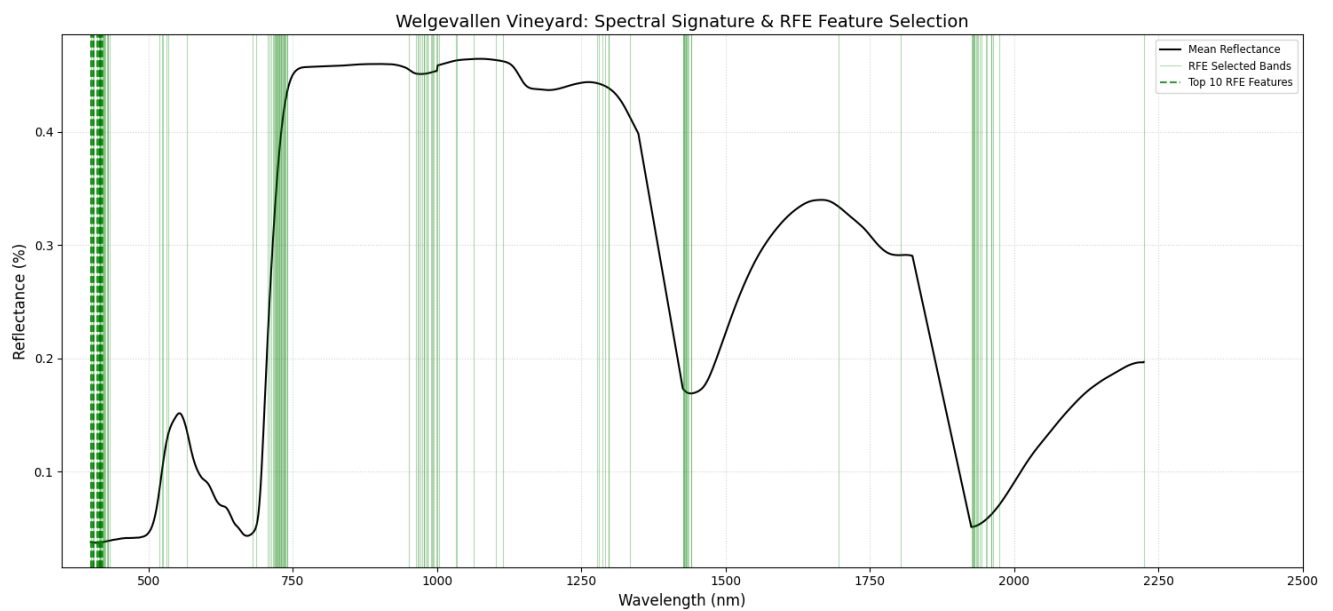


Figure 11: Selected Features using RFE Wrapper feature selection method.

Figure 11 illustrates the features selected by the RFE, which exhibits a wider coverage of the electromagnetic(EM) spectrum than the Boruta in Figure 8. With the only a dense set of features over the Red-Edge which is expected as this region experiences shifts due to the concentration of chlorophyll. As the RFE attempts to select features that are most relevant in showing the statistical relationship between the different cultivar, it doesn't concentrate large amounts of features over similar wavelengths which further decreases redundancy when compared to the Boruta.

Table 3: F1-Score(weighted) and Matthews Correlation Coefficient for the two models using Boruta Feature selection.

	RF RFE	XGB RFE
F1-Score (Weighted)	0,573149	0,625568
MCC	0,487393	0,555553

Table 3 indicates a substantial increase in the F1-score of the XGBoost using RFE feature selection method from the Boruta method. The Random Forest exhibits a slight increase from the Boruta feature selection, this indicates that the features selected by the RFE were more significant to the discrimination of cultivar while reducing the dimensionality of the data. This increase in performance of the RFE is likely due to the focusing of features around the Red-Edge as this region is able to distinguish between the different concentration of chlorophyll which separates the different cultivar. Overall the XGBoost with Boruta outperformed the Random Forest tested on the two different feature selection methods. This further highlights the performance of feature selection methods used on random forest classifiers and the subsetted feature selection used on another machine learning classifier as demonstrated in previous literature (Löw et al., 2013) due to the increase in F1-score.

3.3 HYPERPARAMETER OPTIMISATION AND FEATURE TRANSFORMATION.

After the Savitzky-Golay filter was applied to the model, a PCA feature transformation was performed on the data hyperspectral data. The PCA identified 53 principle components

The use of a grid search to optimise the XGBoost model based on the F1 score found the following hyperparameters; learning_rate: 0.2, max_depth: 3, n_estimators: 1000.

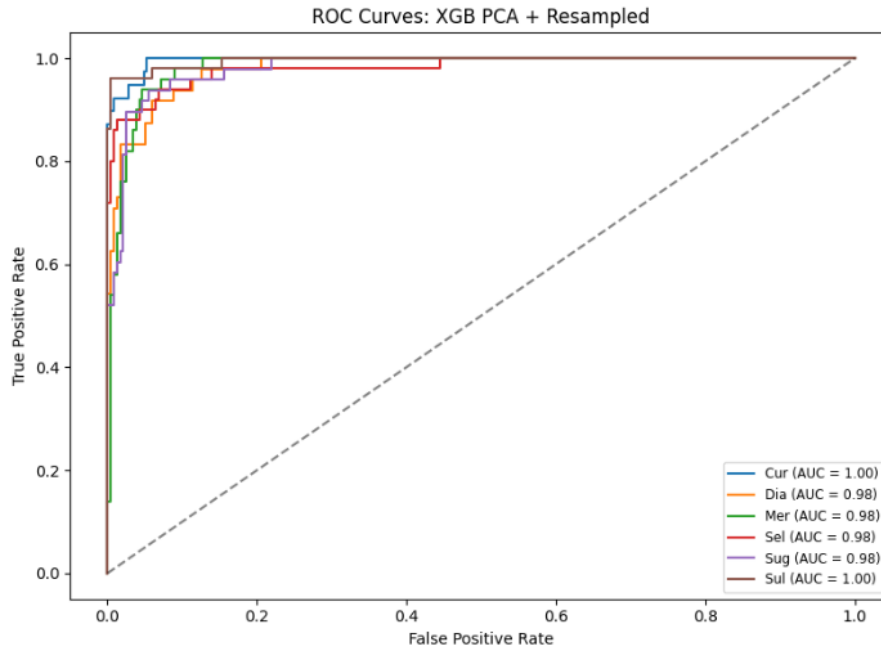


Figure 12: ROC curves and AUC scores of XGBoost final model.

Figure 12 indicates a large true positive rate compared to a false positive rate which is depicted by the ROC curve bending away from the baseline toward the top left. This is an optimal ROC curve for all cultivar as they are almost flush with the top left of the graph. The AUC values of the all cultivar are near the value of 1, this shows a high separability power of the model to distinguish the different cultivar with almost zero overlap between spectral signatures of cultivar.

The final model gathered a training and test accuracy of 1 and 0.88 respectively with an accuracy gap of 0.12. This is a clear indication that overfitting is present in the model, as the ML model has memorised 100% of the data. This overfitting is likely the result of the hyper parameter optimisation where the model was not penalised for the complexity of the classifier.

Table 4: F1 and MCC scores for the final model.

	Final Model
F1-Score (Weighted)	0.884507
MCC	0.861945

The F1-score in Table 4 shows that the final XGBoost model performed significantly well at distinguishing the different cultivar. The MCC score of 0.86 indicates that the model does not randomly assign classes, rather it nearly correctly predicted each instance of a class

Overall the final model performed well, largely due to the hyperparameter optimisation and PCA feature transformation. The hyperparameter optimisation, through the use of a grid search, located optimal hyperparameters that improved the model performance. The PCA was able to transform bands into a smaller subset that allowed the model to accurately predict each cultivar of grapevine. Conversely there was overfitting in the model which gathered a training accuracy of 1, this is likely due to a data leak within the work flow, or the model was not penalised enough on its complexity. This penalisation of model complexity was proven shown to be effective in previous literature (Gutiérrez et al., 2018) as Gutiérrez et al. introduced a penalty parameter of 1.0 that maintained accuracies of 0.99 but did not overfit the model. The Savitzky-Golay filter allowed the PCA to be applied to the full data set with a reduction in the noise inherent to hyperspectral imagery.

3.4 TEMPORAL TRANSFERABILITY

The same model that was trained on the December dataset was tested on the January dataset, this determines the temporal transferability of the model.

The January dataset achieved a F1-score of 0.31 and a MCC score of 0.25, this indicates that the classification of the different grapevine cultivar is near random with low recall and precision.

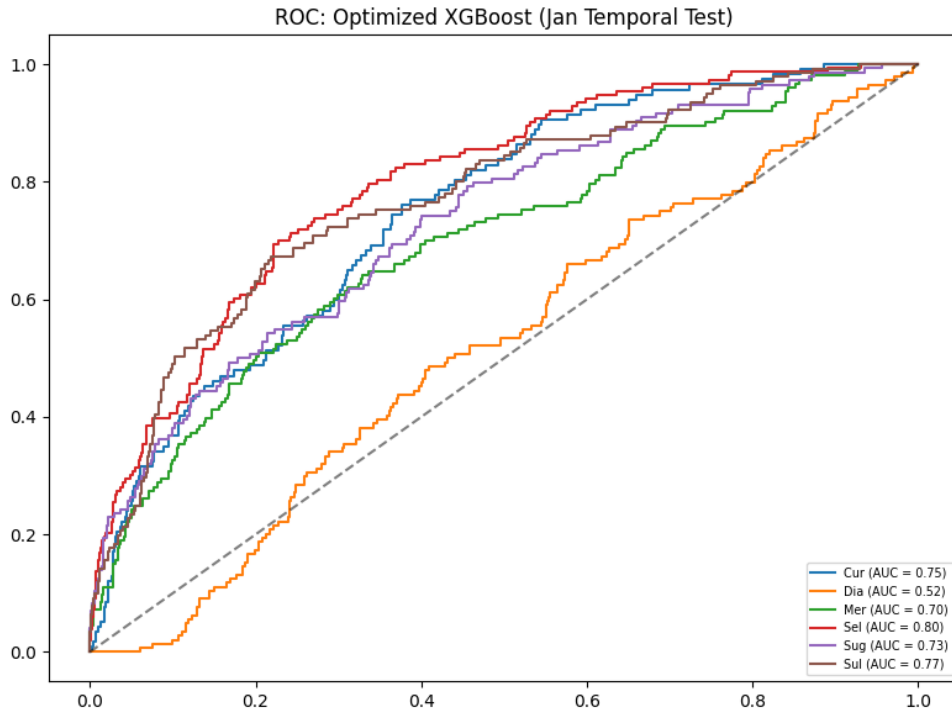


Figure 13: ROC curve and AUC scores of the Temporal transferability test on January data.

The curves shown in Figure 13 are slightly bent toward the top left but follow the baseline. Which further indicates that the classification of the different grapevines is near random.

Overall the temporal transferability of the model is unsuccessful, this is due to the high overfitting of the model to the training data. The difference in the phenological characteristics of grapevines between December and January is large as the nutrients in the leaves are relocated to the ripening grapes. This creates a shift in the spectral signature of December to January data.

4. CONCLUSION

This study aimed to create a machine learning hyperspectral remote sensing framework for the discrimination of six different grapevines at the Welgevallen farm in Stellenbosch. This study successfully compared the baseline of the XGBoost to the Random Forest, where the XGBoost outperformed the Random Forest, with a F1-score of 0.61, on the full hyperspectral dataset. The addition of RFE feature selection, PCA feature transformation, and Savitzky-Golay filter drastically improved the discriminative performance of the selected XGBoost model.

The feature selection highlighted significant wavelengths of the Electromagnetic spectrum, these portions of the spectrum were selected using the RFE and Boruta wrappers. These feature selection methods both highlighted the Red-Edge and SWIR as powerful determining variables in the distinguishability of the different grapevines. Sultana and Currant grapevines exhibited the most unique spectral signature as they maintained high class accuracies through different machine learning algorithms.

The critical challenge encountered in the final XGBoost was the high training accuracy of 1, which indicated that there was overfitting present in the model. This was the likely result of not penalising the model for complexity. This overfitting directly effected the results of the temporal transferability of the model, as testing the model on January data resulted in a F1- score of 0.31.

Overall, while the model proved accurate in discriminating same month, specifically December, grapevine cultivar. For future research, multi temporal data should be incorporated into the training data, this would possibly account for the shift in phenological characteristics of the grapevines.

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